

06_Python → PostgreSQL Integration

Objective

The objective of this phase was to migrate the cleaned and feature-engineered Olist datasets from the Python environment into PostgreSQL for advanced SQL analysis and business intelligence reporting.

Only validated, analysis-ready datasets were transferred to ensure consistency and reliability during downstream analytical activities.

Integration Workflow

The following workflow was implemented:

Raw CSV Files

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Python Data Inspection

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Data Cleaning & Preprocessing

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Feature Engineering

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Python Exploratory Data Analysis

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PostgreSQL Integration

↓

SQL Analysis

↓

Power BI Dashboard Development

Database Environment

Item	Value
Database Name	olist_ecommerce
Database Platform	PostgreSQL
Integration Method	Python to PostgreSQL
Connection Layer	SQLAlchemy & Psycopg2

Datasets Migrated

The following cleaned datasets were successfully migrated into PostgreSQL:

Table Name

customers

orders

payments

reviews

products

sellers

order_items

geolocation

category_translation

Engineered Features Migrated

The following engineered features were retained during migration and made available for SQL analysis:

- approval_days
- carrier_days
- delivery_days
- total_fulfillment_days
- delivery_delay_days

- purchase_year
- purchase_month
- purchase_quarter
- purchase_weekday

These features enhance operational, logistics, customer experience, and trend analysis capabilities within SQL and Power BI.

Validation Checks

Post-migration validation was performed to ensure data integrity.

Validation activities included:

- Database connection verification
- Table creation verification
- Row count validation
- Engineered feature validation
- Table structure review

All validation checks were completed successfully.

Outcome

All cleaned and feature-engineered datasets were successfully migrated into PostgreSQL.

The database environment now contains validated, analysis-ready data prepared for:

- SQL Exploratory Data Analysis
- KPI Development
- Business Insight Generation
- Hypothesis Testing
- Power BI Dashboard Development

Phase Completion

The Python phase concluded with successful PostgreSQL integration and database validation.

Deliverables Completed

Data Inspection & Quality Audit

Referential Integrity Validation

Data Cleaning & Preprocessing

Feature Engineering

Python Exploratory Data Analysis

PostgreSQL Integration

The project is now ready to transition into the SQL Analysis phase.